Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18AI7DE42]

# SEVENTH SEMESTER B. TECH EXAMINATION, DECEMBER-2021

## NOSQL DATABASES

**Time: 3 Hours** **Maximum Marks: 70**

**Instructions:**
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

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|---|---|---|---|
| **1.** | **a.** | Explain NoSQL system used by any company in the industry and demonstrate how NoSQL technologies are used to solve a difficult analytical problem that traditional SQL database could not. | **07** |
| | **b.** | Describe the transactional ACID properties and the BASE properties. Compare the two sets of properties and explain why BASE properties are used in some systems. | **04** |
| | **c.** | Discuss the need for both NoSQL and relational databases in enterprise data management. | **03** |
| **2.** | **a.** | There are three primary concerns you must balance when choosing a data management system: consistency, availability, and partition tolerance. According to the CAP Theorem, you can only pick two. So, how does this all relate to NoSQL systems? | **07** |
| | **b.** | A single application that uses different types of data needs to use multiple databases behind that application. Suggest suitable database and justify with examples. | **07** |
| **3.** | **a.** | Write down at least three syntax rules for JSON objects. Create a sample document for a small appliance with the following attributes: appliance ID, name, description, height, width, length, and shipping weight. Use the JSON format. | **07** |
| | **b.** | List any four motivating factors for database designers and other IT professionals to develop and use Document based NoSQL databases. | **07** |

**OR**

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| **4.** | **a.** | Describe a situation where it would make sense to have many indexes on your document collections. | **07** |
| | **b.** | Define sharding. | **02** |
| | **c.** | List out the components of a MongoDB Sharded Cluster. | **03** |
| | **d.** | Can you change the shard key after a collection is sharded? Justify your answer. | **02** |
| **5.** | **a.** | Differentiate between key value store and document-oriented database. | **03** |

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|---|---|---|---|
| | b. | Describe a way of constructing keys that captures some information about entities and attribute types. | 04 |
| | c. | Why would you want to receive a response from more than one replica when reading a value from a key-value data store? | 03 |
| | d. | Describe the Time to Live (TTL) key pattern. | 04 |
| | | **OR** | |
| 6. | a. | Define the use cases where key-value databases can be used along with examples | 07 |
| | b. | List out any four common features of key-value databases. Explain with an example. | 07 |
| 7. | a. | Give an example of a business application that would use multiple types of edges (relations) between vertices with explanation. | 07 |
| | b. | List out at least three examples in which you can use graphs to model the domains. | 03 |
| | c. | Describe two reasons for choosing a graph database for your application. | 04 |
| | | **OR** | |
| 8. | a. | Illustrate the difference between a declarative and a traversal query language for graph databases. | 05 |
| | b. | Which type of edge should be used for a nonsymmetrical relation, a directed or undirected edge? | 05 |
| | c. | List out the benefit of mapping domain-specific queries into graph-specific queries. | 04 |